Project Report

Text Data Curation Pipeline using NeMo Concepts

Cleaning and optimizing conversational datasets for AI applications

THAPAR INSTITUTE
OF ENGINEERING & TECHNOLOGY
(Deemed to be University)

Advit Budhiraja - 1024240130
Batch- 2X14

INDEX

LINK TO COLAB NOTEBOOK
https://colab.research.google.com/drive/11v5xXnpUKG8Dfgg3WfQ8q5MS2gEvrSvr?usp=sharing

1. Problem Statement, Objective, and Applications

Problem Statement

Raw textual datasets, especially conversational data, often contain:

- Noise and inconsistencies (formatting issues, casing, spacing)
- Duplicate entries
- Low-quality or meaningless text

These issues negatively impact AI models by:

- Reducing training efficiency
- Lowering prediction accuracy
- Increasing computational cost

Objective

The objective of this project is to design and implement a data curation pipeline that:

- Cleans raw text data
- Filters out low-quality entries
- Removes duplicate content
- Produces a structured, high-quality dataset suitable for AI/ML tasks

Applications

The curated dataset can be used in:

- Chatbot development
- Natural Language Processing (NLP) models
- Conversational AI systems
- Large Language Model (LLM) fine-tuning

2. Pipeline Overview

The project follows a structured pipeline:

1. Input raw dataset (text format)
2. Convert data into JSONL format
3. Apply text cleaning
4. Apply filtering rules
5. Perform deduplication
6. Save cleaned dataset
7. Evaluate improvements using metrics and visualization

The final output is a **clean, structured JSONL dataset** ready for AI training.

3. Cleaning, Filtering, and Deduplication

Text Cleaning

The dataset is normalized using:

- Conversion to lowercase
- Removal of extra spaces
- Fixing Unicode characters (e.g., smart quotes)
- Basic punctuation correction

This ensures **consistency and readability**

Filtering

Low-quality data is removed using rules such as:

- Eliminating very short text entries
- Removing numeric-only content

This helps in retaining only **meaningful textual data**

Deduplication

- Duplicate sentences are identified and removed
- Only unique entries are retained

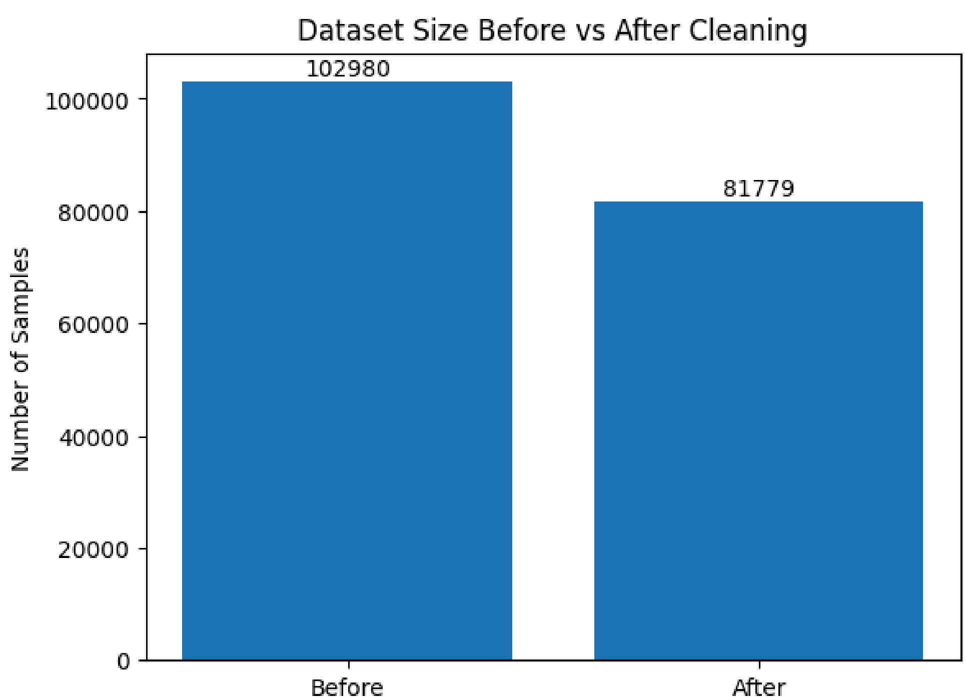
This improves:

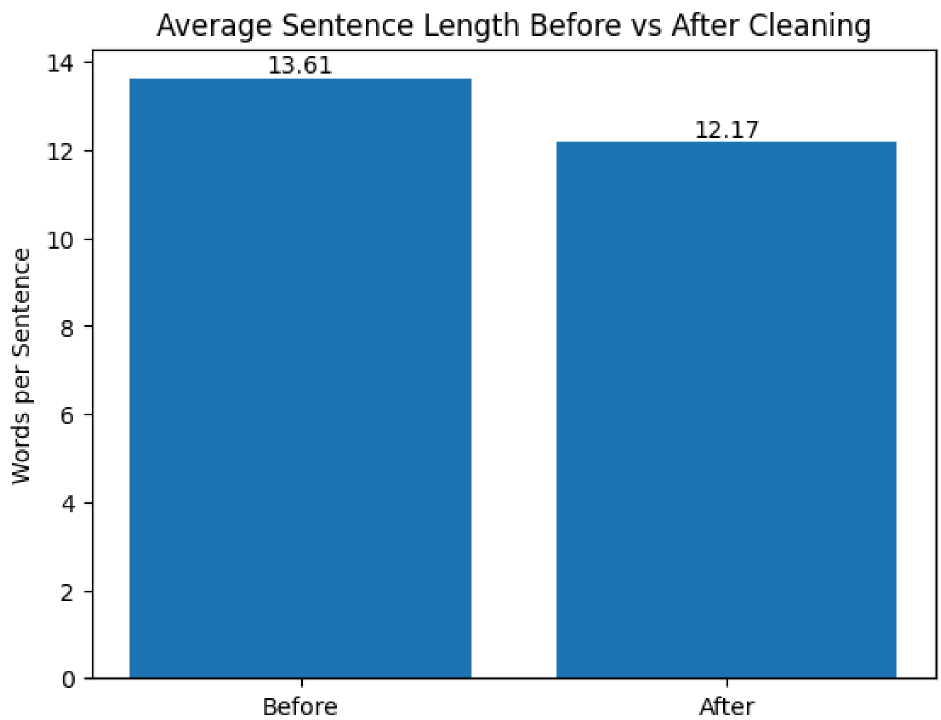
- Dataset diversity
- Model generalization

Results

- Dataset reduced from **5000 → 4680 samples (~6.4% reduction)**
- Average sentence length slightly decreased:
    - Before: **13.61 words**
    - After: **12.17 words**

Interpretation:
The decrease indicates removal of duplicate conversational patterns (often longer), while preserving natural sentence diversity.





4. Tech Stack and Challenges Faced

Tech Stack

- Python
- JSON / JSONL
- Regular Expressions (Regex)
- Matplotlib (for visualization)
- NVIDIA NeMo Curator (data curation framework)

Challenges Faced

1. **Platform Limitation**
   - NeMo Curator only supports Linux
   - Not compatible with macOS
2. **Version Compatibility Issues**

    - Library API mismatches in different environments
    - Import errors in Colab
3. **Dataset Loading Issues**
    - Some datasets deprecated or unsupported
    - Required manual dataset handling

Solution

Instead of relying on unstable libraries:
A **custom data curation pipeline** was implemented manually using Python

This ensured:

- Full control
- Stability
- Better understanding of the process

5. Future Improvements

The pipeline can be enhanced further by adding:

- **Word-based filtering** for better quality control
- **Profanity filtering** to remove inappropriate content
- **Language detection** for multilingual datasets
- **Semantic deduplication** (removing similar meaning sentences)
- **Quality scoring system** to rank dataset entries

These improvements would make the system closer to industry-grade data curation tools

6. Conclusion

This project successfully demonstrates the importance of **data preprocessing in AI pipelines**.

Key achievements:

- Built a complete data curation pipeline
- Improved dataset quality
- Reduced redundancy
- Maintained natural conversational structure

High-quality data is the foundation of effective AI systems